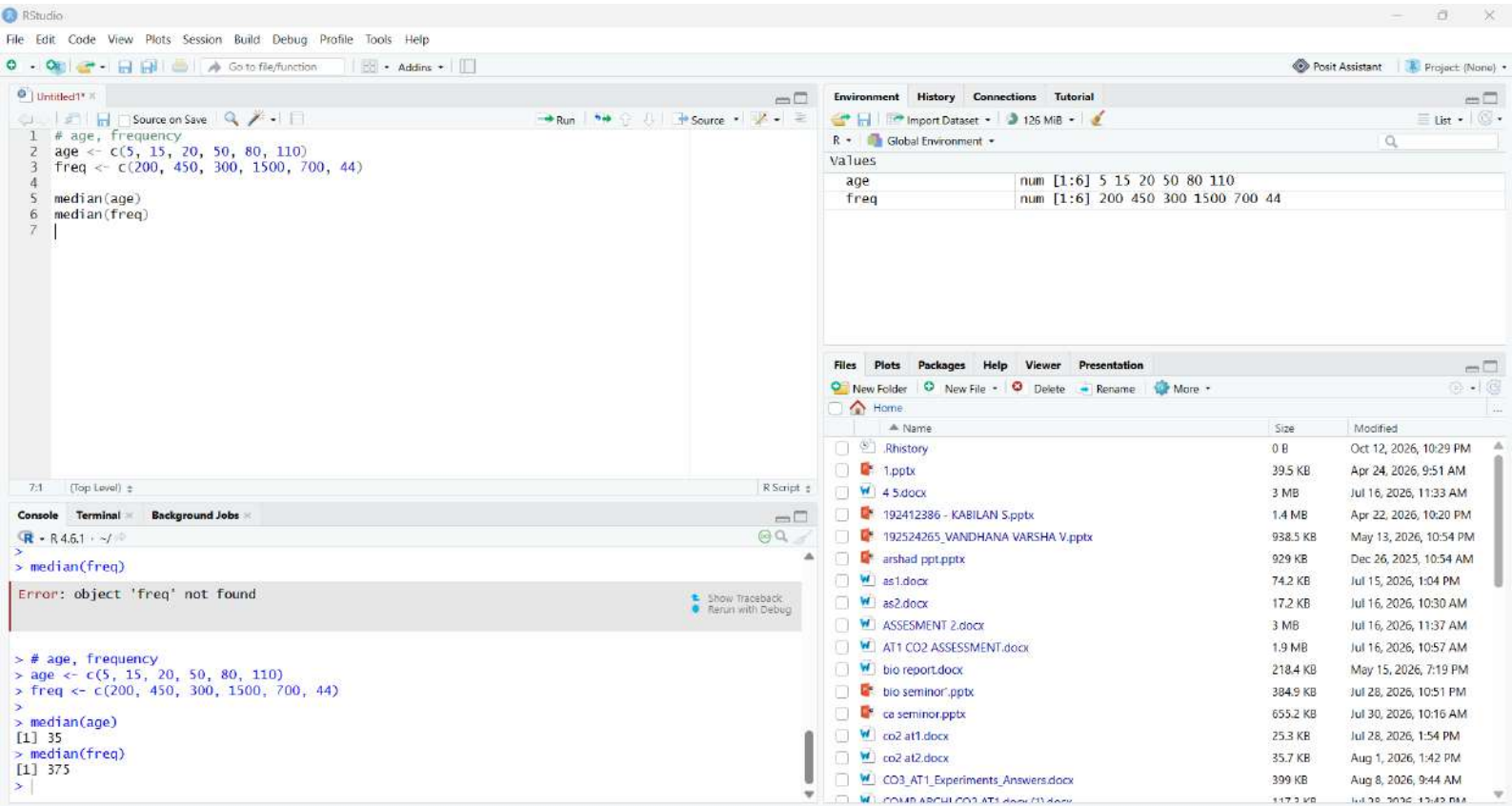
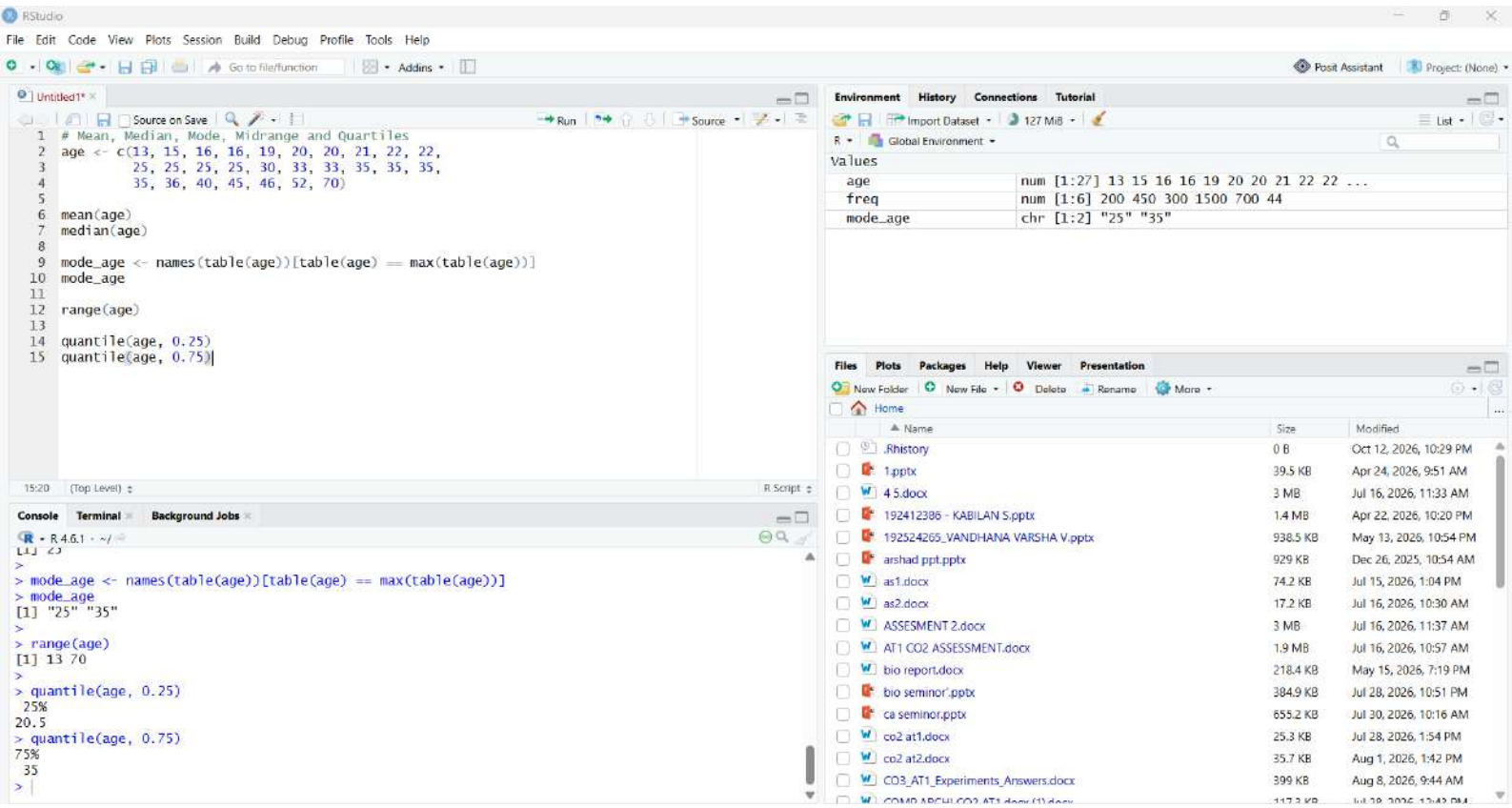
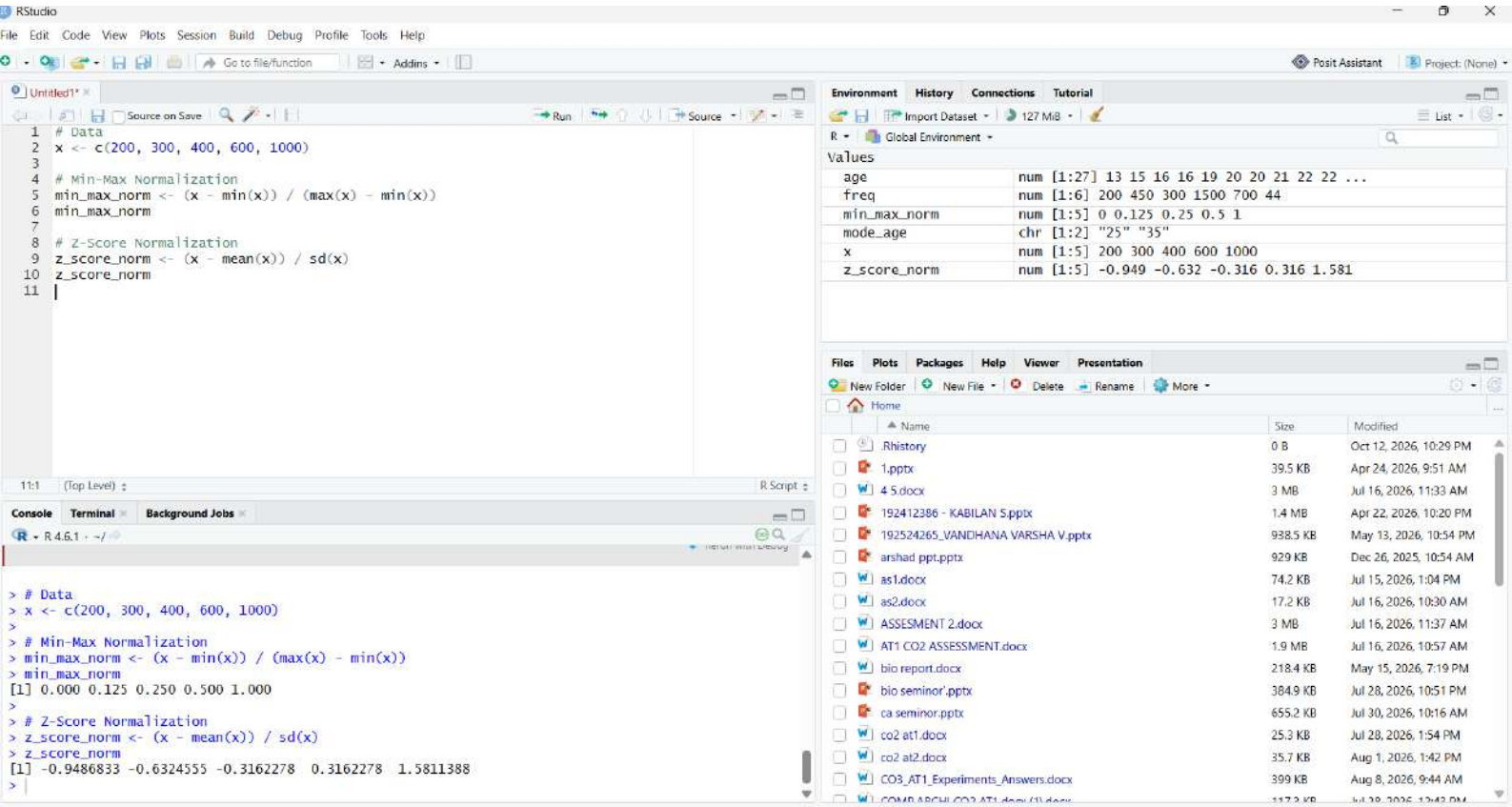
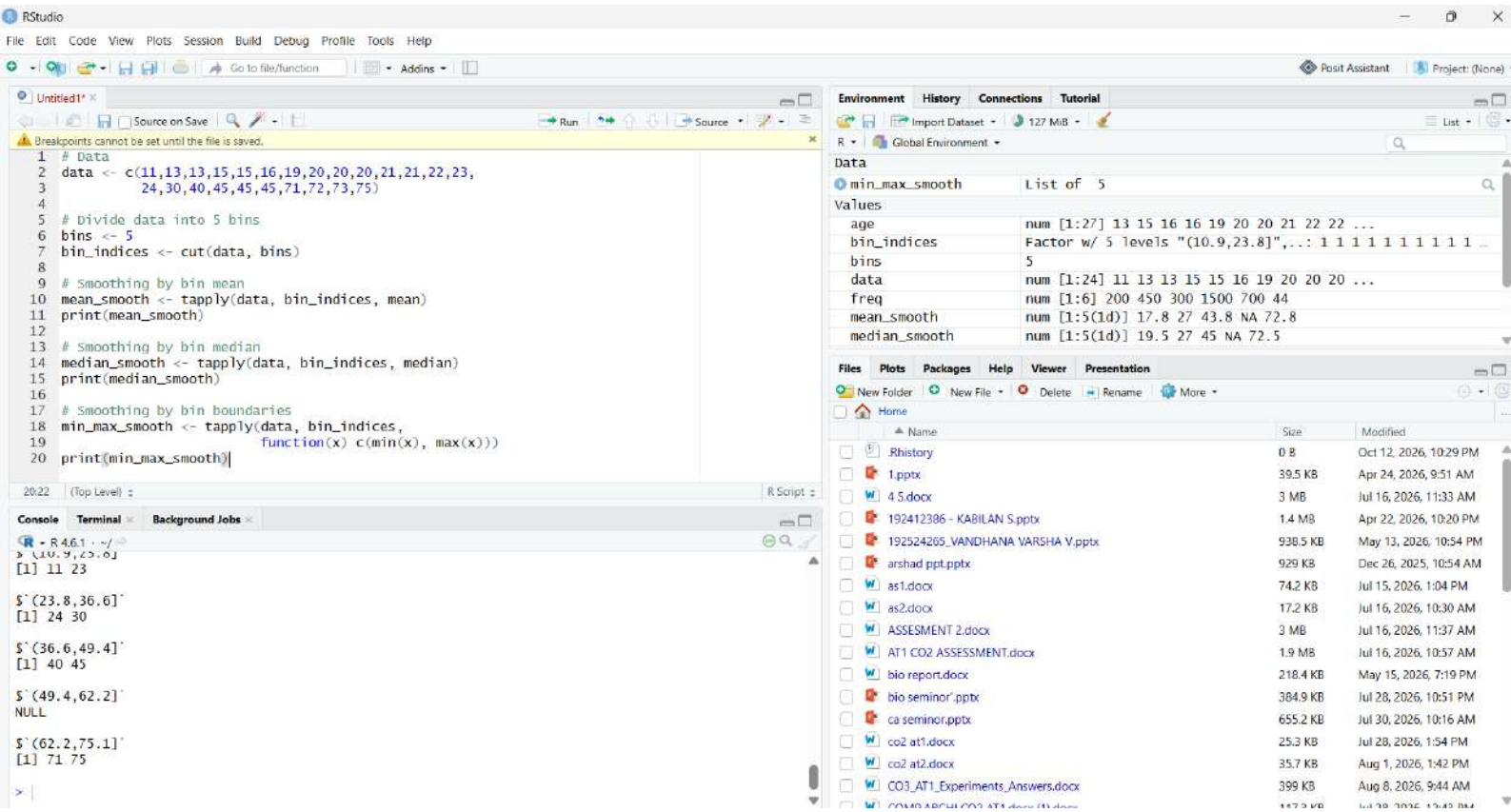

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
1 # age, frequency
2 age <- c(5, 15, 20, 50, 80, 110)
3 freq <- c(200, 450, 300, 1500, 700, 44)
4
5 median(age)
6 median(freq)
7
Environment History Connections Tutorial
Import Dataset
Global Environment
Values
age num [1:6] 5 15 20 50 80 110
freq num [1:6] 200 450 300 1500 700 44
Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Home
Name Size Modified
.Rhistory 0 B Oct 12, 2026, 10:29 PM
1.pptx 39.5 KB Apr 24, 2026, 9:51 AM
4 5.docx 3 MB Jul 16, 2026, 11:33 AM
192412386 - KABILAN S.pptx 1.4 MB Apr 22, 2026, 10:20 PM
192524265_VANDHANA VARSHA V.pptx 938.5 KB May 13, 2026, 10:54 PM
arshad ppt.pptx 929 KB Dec 26, 2025, 10:54 AM
as1.docx 74.2 KB Jul 15, 2026, 1:04 PM
as2.docx 17.2 KB Jul 16, 2026, 10:30 AM
ASSESMENT 2.docx 3 MB Jul 16, 2026, 11:37 AM
AT1 CO2 ASSESSMENT.docx 1.9 MB Jul 16, 2026, 10:57 AM
bio report.docx 218.4 KB May 15, 2026, 7:19 PM
bio seminor.pptx 384.9 KB Jul 28, 2026, 10:51 PM
ce seminor.pptx 655.2 KB Jul 30, 2026, 10:16 AM
co2 at1.docx 25.3 KB Jul 28, 2026, 1:54 PM
co2 at2.docx 35.7 KB Aug 1, 2026, 1:42 PM
CO3_AT1_Experiments_Answers.docx 399 KB Aug 8, 2026, 9:44 AM
Console Terminal Background Jobs
R 4.6.1 · ~/
> median(freq)
Error: object 'freq' not found
Show Traceback
Rerun with Debug
> # age, frequency
> age <- c(5, 15, 20, 50, 80, 110)
> freq <- c(200, 450, 300, 1500, 700, 44)
> median(age)
[1] 35
> median(freq)
[1] 375


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
Mean, Median, Mode, Midrange and Quartiles
age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22, 22,
25, 25, 25, 25, 30, 33, 33, 35, 35, 35,
35, 36, 40, 45, 46, 52, 70)
mean(age)
median(age)
mode_age <- names(table(age))[table(age) == max(table(age))]
mode_age
range(age)
quantile(age, 0.25)
quantile(age, 0.75)
15:20 (Top Level) R Script
Console Terminal Background Jobs
R 4.4.1 · ~/
> mode_age <- names(table(age))[table(age) == max(table(age))]
> mode_age
[1] "25" "35"
> range(age)
[1] 13 70
> quantile(age, 0.25)
25%
20.5
> quantile(age, 0.75)
75%
35
Environment History Connections Tutorial
Import Dataset 127 MiB
Global Environment
Values
age num [1:27] 13 15 16 16 19 20 20 21 22 22 ...
freq num [1:6] 200 450 300 1500 700 44
mode_age chr [1:2] "25" "35"
Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Home
Name Size Modified
.Rhistory 0 B Oct 12, 2026, 10:29 PM
1.pptx 39.5 KB Apr 24, 2026, 9:51 AM
4 5.docx 3 MB Jul 16, 2026, 11:33 AM
192412386 - KABILAN S.pptx 1.4 MB Apr 22, 2026, 10:20 PM
192524265_VANDHANA VARSHA V.pptx 938.5 KB May 13, 2026, 10:54 PM
arshad ppt.pptx 929 KB Dec 26, 2025, 10:54 AM
as1.docx 74.2 KB Jul 15, 2026, 1:04 PM
as2.docx 17.2 KB Jul 16, 2026, 10:30 AM
ASSESMENT 2.docx 3 MB Jul 16, 2026, 11:37 AM
AT1 CO2 ASSESSMENT.docx 1.9 MB Jul 16, 2026, 10:57 AM
bio report.docx 218.4 KB May 15, 2026, 7:19 PM
bio seminor'.pptx 384.9 KB Jul 28, 2026, 10:51 PM
ca seminor.pptx 655.2 KB Jul 30, 2026, 10:16 AM
co2 at1.docx 25.3 KB Jul 28, 2026, 1:54 PM
co2 at2.docx 35.7 KB Aug 1, 2026, 1:42 PM
CO3_AT1_Experiments_Answers.docx 399 KB Aug 8, 2026, 9:44 AM


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
1 # Data
2 x <- c(200, 300, 400, 600, 1000)
3
4 # Min-Max Normalization
5 min_max_norm <- (x - min(x)) / (max(x) - min(x))
6 min_max_norm
7
8 # Z-Score Normalization
9 z_score_norm <- (x - mean(x)) / sd(x)
10 z_score_norm
11
11:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> # Data
> x <- c(200, 300, 400, 600, 1000)
>
> # Min-Max Normalization
> min_max_norm <- (x - min(x)) / (max(x) - min(x))
> min_max_norm
[1] 0.000 0.125 0.250 0.500 1.000
>
> # Z-Score Normalization
> z_score_norm <- (x - mean(x)) / sd(x)
> z_score_norm
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
>
Environment
History
Connections
Tutorial
Import Dataset
127 MiB
List
R
Global Environment
Values
age num [1:27] 13 15 16 16 19 20 20 21 22 22 ...
freq num [1:6] 200 450 300 1500 700 44
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode_age chr [1:2] "25" "35"
x num [1:5] 200 300 400 600 1000
z_score_norm num [1:5] -0.949 -0.632 -0.316 0.316 1.581
Files
Plots
Packages
Help
Viewer
Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
.Rhistory 0 B Oct 12, 2026, 10:29 PM
1.pptx 39.5 KB Apr 24, 2026, 9:51 AM
4 5.docx 3 MB Jul 16, 2026, 11:33 AM
192412386 - KABILAN S.pptx 1.4 MB Apr 22, 2026, 10:20 PM
192524265_VANDHANA VARSHA V.pptx 938.5 KB May 13, 2026, 10:54 PM
arshad ppt.pptx 929 KB Dec 26, 2025, 10:54 AM
as1.docx 74.2 KB Jul 15, 2026, 1:04 PM
as2.docx 17.2 KB Jul 16, 2026, 10:30 AM
ASSESMENT 2.docx 3 MB Jul 16, 2026, 11:37 AM
AT1 CO2 ASSESSMENT.docx 1.9 MB Jul 16, 2026, 10:57 AM
bio report.docx 218.4 KB May 15, 2026, 7:19 PM
bio seminor'.pptx 384.9 KB Jul 28, 2026, 10:51 PM
ca seminor.pptx 655.2 KB Jul 30, 2026, 10:16 AM
co2 at1.docx 25.3 KB Jul 28, 2026, 1:54 PM
co2 at2.docx 35.7 KB Aug 1, 2026, 1:42 PM
CO3_AT1_Experiments_Answers.docx 399 KB Aug 8, 2026, 9:44 AM


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
Breakpoints cannot be set until the file is saved.
Data
data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,
24,30,40,45,45,45,71,72,73,75)
Divide data into 5 bins
bins <- 5
bin_indices <- cut(data, bins)
Smoothing by bin mean
mean_smooth <- tapply(data, bin_indices, mean)
print(mean_smooth)
Smoothing by bin median
median_smooth <- tapply(data, bin_indices, median)
print(median_smooth)
Smoothing by bin boundaries
min_max_smooth <- tapply(data, bin_indices,
function(x) c(min(x), max(x)))
print(min_max_smooth)
20:22 (Top Level) R Script
Console Terminal Background Jobs
R 4.4.1 · ~/
[1] 11 23
$`(23.8,36.6]`
[1] 24 30
$`(36.6,49.4]`
[1] 40 45
$`(49.4,62.2]`
NULL
$`(62.2,75.1]`
[1] 71 75
Environment History Connections Tutorial
Import Dataset
127 MiB
List
R
Global Environment
Data
min_max_smooth List of 5
Values
age num [1:27] 13 15 16 16 19 20 20 21 22 22 ...
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1 1 1 1 1 1 ...
bins 5
data num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
freq num [1:6] 200 450 300 1500 700 44
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
.Rhistory 0 B Oct 12, 2026, 10:29 PM
1.pptx 39.5 KB Apr 24, 2026, 9:51 AM
4 5.docx 3 MB Jul 16, 2026, 11:33 AM
192412386 - KABILAN S.pptx 1.4 MB Apr 22, 2026, 10:20 PM
192524265_VANDHANA VARSHA V.pptx 938.5 KB May 13, 2026, 10:54 PM
arshad ppt.pptx 929 KB Dec 26, 2025, 10:54 AM
as1.docx 74.2 KB Jul 15, 2026, 1:04 PM
as2.docx 17.2 KB Jul 16, 2026, 10:30 AM
ASSESMENT 2.docx 3 MB Jul 16, 2026, 11:37 AM
AT1 CO2 ASSESSMENT.docx 1.9 MB Jul 16, 2026, 10:57 AM
bio report.docx 218.4 KB May 15, 2026, 7:19 PM
bio seminor'.pptx 384.9 KB Jul 28, 2026, 10:51 PM
ca seminor.pptx 655.2 KB Jul 30, 2026, 10:16 AM
co2 at1.docx 25.3 KB Jul 28, 2026, 1:54 PM
co2 at2.docx 35.7 KB Aug 1, 2026, 1:42 PM
CO3_AT1_Experiments_Answers.docx 399 KB Aug 8, 2026, 9:44 AM

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Untitled1*

Source on Save | Run | Source

Breakpoints cannot be set until the file is saved.

```
4
5  fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
6           34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7)
7
8  # Mean, median and standard deviation
9  mean(age)
10 median(age)
11 sd(age)
12
13 mean(fat)
14 median(fat)
15 sd(fat)
16
17 # Boxplot
18 boxplot(age, fat)
19
20 # Scatterplot
21 scatter.smooth(age, fat)
22
23 # Q-Q plot
24 qqplot(age, fat)
```

7:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.4.1 · ~/

```
> # Mean, median and standard deviation
> mean(age)
[1] 46.44444
> median(age)
[1] 51
> sd(age)
[1] 13.21862
>
> mean(fat)
[1] 28.78333
> median(fat)
[1] 30.7
> sd(fat)
[1] 9.254395
>
```

Environment History Connections Tutorial

Import Dataset | 128 MiB | List

R | Global Environment

Data

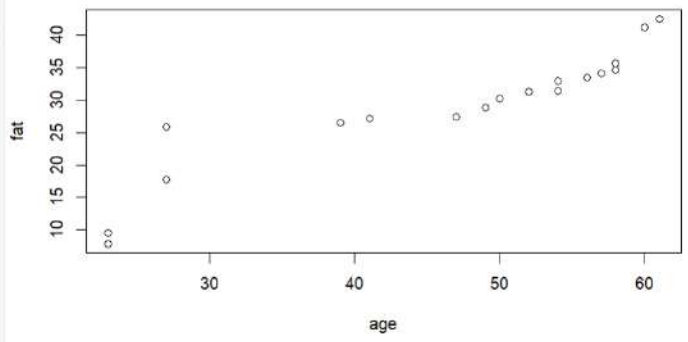
min_max_smooth	List of 5

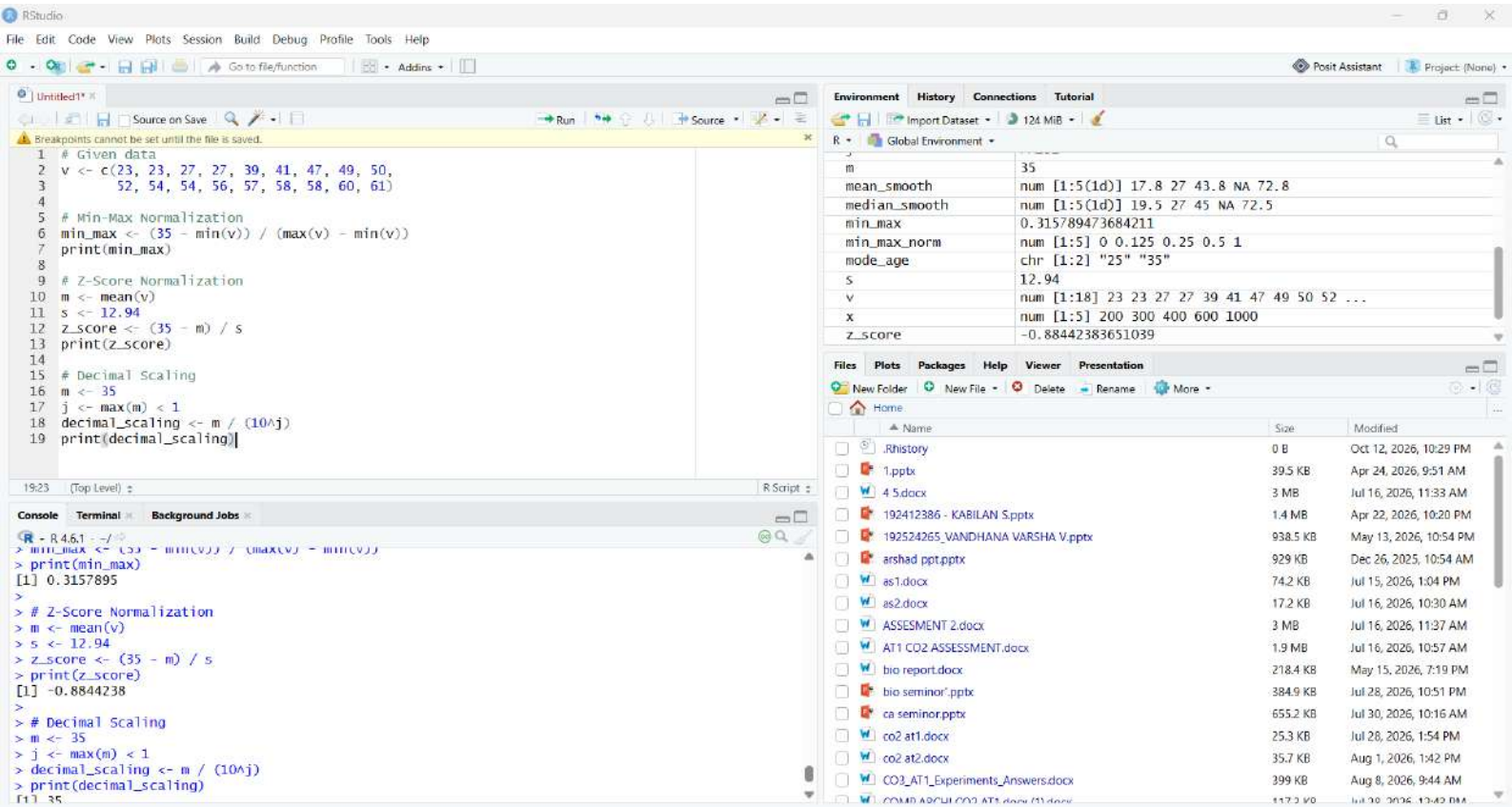
Values

age	num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
bin_indices	Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1 1 1 1 1 1 ...
bins	5
data	num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
fat	num [1:18] 9.5 26.5 7.8 17.8 31.4 25.9 27.4 27.2 31.2 34....
freq	num [1:6] 200 450 300 1500 700 44
mean_smooth	num [1:5(1d)] 17.8 27 43.8 NA 72.8

Files Plots Packages Help Viewer Presentation

Zoom | Export | Publish




RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
Breakpoints cannot be set until the file is saved.
Given data
v <- c(23, 23, 27, 27, 39, 41, 47, 49, 50,
52, 54, 54, 56, 57, 58, 58, 60, 61)
Min-Max Normalization
min_max <- (35 - min(v)) / (max(v) - min(v))
print(min_max)
Z-Score Normalization
m <- mean(v)
s <- 12.94
z_score <- (35 - m) / s
print(z_score)
Decimal Scaling
m <- 35
j <- max(m) < 1
decimal_scaling <- m / (10^j)
print(decimal_scaling)
19:23 (Top Level) R Script
Console Terminal Background Jobs
R 4.4.1 · ~/
> print(min_max)
[1] 0.3157895
> # Z-Score Normalization
> m <- mean(v)
> s <- 12.94
> z_score <- (35 - m) / s
> print(z_score)
[1] -0.8844238
> # Decimal Scaling
> m <- 35
> j <- max(m) < 1
> decimal_scaling <- m / (10^j)
> print(decimal_scaling)
[1] 35
Environment History Connections Tutorial
Import Dataset 124 MiB
R Global Environment
m 35
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode_age chr [1:2] "25" "35"
s 12.94
v num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x num [1:5] 200 300 400 600 1000
z_score -0.88442383651039
Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Home
Name Size Modified
.Rhistory 0 B Oct 12, 2026, 10:29 PM
1.pptx 39.5 KB Apr 24, 2026, 9:51 AM
4 5.docx 3 MB Jul 16, 2026, 11:33 AM
192412386 - KABILAN S.pptx 1.4 MB Apr 22, 2026, 10:20 PM
192524265_VANDHANA VARSHA V.pptx 938.5 KB May 13, 2026, 10:54 PM
arshad ppt.pptx 929 KB Dec 26, 2025, 10:54 AM
as1.docx 74.2 KB Jul 15, 2026, 1:04 PM
as2.docx 17.2 KB Jul 16, 2026, 10:30 AM
ASSESMENT 2.docx 3 MB Jul 16, 2026, 11:37 AM
AT1 CO2 ASSESSMENT.docx 1.9 MB Jul 16, 2026, 10:57 AM
bio report.docx 218.4 KB May 15, 2026, 7:19 PM
bio seminor'.pptx 384.9 KB Jul 28, 2026, 10:51 PM
ca seminor.pptx 655.2 KB Jul 30, 2026, 10:16 AM
co2 at1.docx 25.3 KB Jul 28, 2026, 1:54 PM
co2 at2.docx 35.7 KB Aug 1, 2026, 1:42 PM
CO3_AT1_Experiments_Answers.docx 399 KB Aug 8, 2026, 9:44 AM

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

Untitled1*

Source on Save | Run | Source

Breakpoints cannot be set until the file is saved.

```
1  # Number of pencils in boxes
2  pencils <- c(9, 25, 23, 12, 11, 6, 7, 8, 9, 10)
3
4  # Mean
5  mean(pencils)
6
7  # Median
8  median(pencils)
9
10 # Mode
11 mode <- names(table(pencils))[table(pencils) == max(table(pencils))]
12 print(mode)
```

12:12 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 · ~/

```
> # Mean
> mean(pencils)
[1] 12
>
> # Median
> median(pencils)
[1] 9.5
>
> # Mode
> mode <- names(table(pencils))[table(pencils) == max(table(pencils))]
> print(mode)
[1] "9"
> source("~/.active-rstudio-document")
[1] "9"
>
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Untitled1*

Source on Save | Run | Source

Breakpoints cannot be set until the file is saved.

```
1  # X and Y values
2  x <- c(4, 1, 5, 7, 10, 2, 50, 25, 90, 36)
3  y <- c(12, 5, 13, 19, 31, 7, 153, 72, 275, 110)
4
5  # Scatter plot
6  scatter.smooth(x, y)
```

6:21 (Top Level) R Script

Console Terminal Background Jobs

R 4.4.1 · ~/

```
> median(pencils)
[1] 9.5
>
> # Mode
> mode <- names(table(pencils))[table(pencils) == max(table(pencils))]
> print(mode)
[1] "9"
> source("~/.active-rstudio-document")
[1] "9"
> # X and Y values
> x <- c(4, 1, 5, 7, 10, 2, 50, 25, 90, 36)
> y <- c(12, 5, 13, 19, 31, 7, 153, 72, 275, 110)
>
> # Scatter plot
> scatter.smooth(x, y)
```

Environment History Connections Tutorial

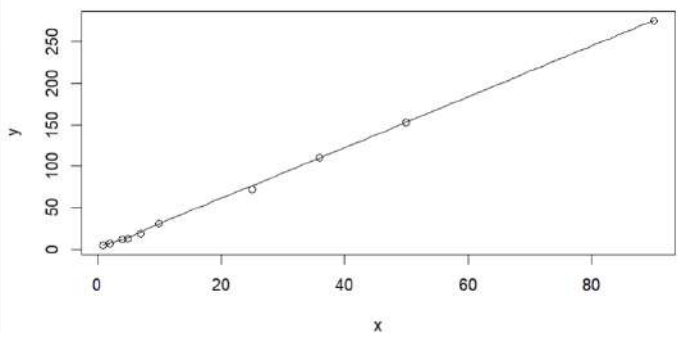
Import Dataset | 125 MiB | List

R | Global Environment

min_max_norm	num [1:5] 0 0.125 0.25 0.5 1
mode	"9"
mode_age	chr [1:2] "25" "35"
pencils	num [1:10] 9 25 23 12 11 6 7 8 9 10
s	12.94
v	num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x	num [1:10] 4 1 5 7 10 2 50 25 90 36
y	num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score	-0.88442383651039
z_score_norm	num [1:5] -0.949 -0.632 -0.316 0.316 1.581

Files Plots Packages Help Viewer Presentation

Zoom | Export | Publish



File Edit Code View Plots Session Build Debug Profile Tools Help

Untitled1*

Source on Save | Run | Source

Breakpoints cannot be set until the file is saved.

```
4
5  # Number of bins
6  num_bins <- 3
7
8  # Equal-frequency partitioning
9  bins_eq_frequency <- cut(marks, breaks = num_bins, labels = FALSE)
10 hist(marks, breaks = num_bins, col = "lightblue",
11      xlab = "Marks", main = "Equal-Width Partitioning")
12
13 # Smoothing by bin mean
14 bin_mean <- tapply(marks, bins_eq_frequency, mean)
15 print(bin_mean)
16
17 # Smoothing by bin median
18 bin_median <- tapply(marks, bins_eq_frequency, median)
19 print(bin_median)
20
21 # Smoothing by bin boundaries
22 bin_boundaries <- tapply(marks, bins_eq_frequency,
23                          function(x) c(min(x), max(x)))
24 print(bin_boundaries)
```

17:26 (Top Level) R Script

Console Terminal Background Jobs

R 4.4.1 · ~/

```
> # Smoothing by bin boundaries
> bin_boundaries <- tapply(marks, bins_eq_frequency,
+                          function(x) c(min(x), max(x)))
> print(bin_boundaries)
$`1`
[1] 50 58

$`2`
[1] 59 65

$`3`
[1] 67 75

>
```

Environment History Connections Tutorial

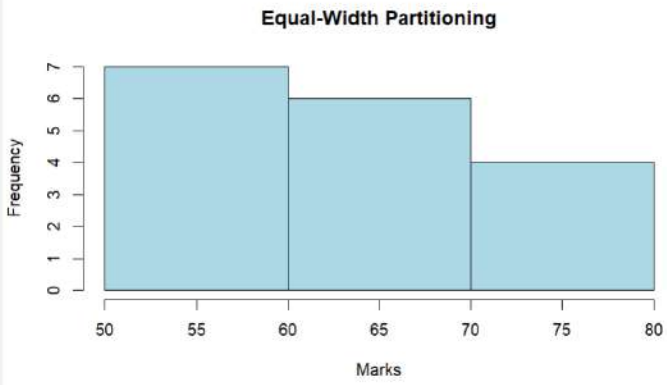
Import Dataset | 125 MiB | List

R | Global Environment

mode	"9"
mode_age	chr [1:2] "25" "35"
num_bins	3
pencils	num [1:10] 9 25 23 12 11 6 7 8 9 10
s	12.94
v	num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x	num [1:10] 4 1 5 7 10 2 50 25 90 36
y	num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score	-0.88442383651039
z_score_norm	num [1:5] -0.949 -0.632 -0.316 0.316 1.581

Files Plots Packages Help Viewer Presentation

Zoom | Export | Publish



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

Untitled1*

Source on Save | Run | Source

Breakpoints cannot be set until the file is saved.

```
1  # Speed data
2  v <- c(78.3, 81.8, 82, 74.2, 83.4, 84.5, 82.9, 77.5, 80.9, 70.6)
3
4  # Interquartile Range
5  IQR(v)
6
7  # Standard Deviation
8  sd(v)
```

7:21 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 · ~/

```
$`3`
[1] 67 75

> # Speed data
> v <- c(78.3, 81.8, 82, 74.2, 83.4, 84.5, 82.9, 77.5, 80.9, 70.6)
>
> # Interquartile Range
> IQR(v)
[1] 4.975
>
> # Standard Deviation
> sd(v)
[1] 4.445835
>
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

Untitled1*

Source on Save | Run | Source

Breakpoints cannot be set until the file is saved.

```
1  age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22, 22,
2           25, 25, 25, 25, 30, 33, 33, 35, 35, 35,
3           35, 36, 40, 45, 46, 52, 70)
4
5  # First Quartile (Q1)
6  quantile(age, 0.25)
7
8  # Third Quartile (Q3)
9  quantile(age, 0.75)
```

9:20 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 · ~/

```
> sd(v)
[1] 4.445835
> age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22, 22,
+          25, 25, 25, 25, 30, 33, 33, 35, 35, 35,
+          35, 36, 40, 45, 46, 52, 70)
>
> # First Quartile (Q1)
> quantile(age, 0.25)
 25%
20.5
>
> # Third Quartile (Q3)
> quantile(age, 0.75)
75%
 35
>
```